

ZAINEB MESSAOUDI

Software Engineering Student

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Portfolio : zainebportfolio.vercel.app

PROFESSIONAL SUMMARY

2nd year Software Engineering student at ESPRIT specializing in Web & Internet Technologies, with experience in full-stack development and applied AI systems through internships and academic projects. Built web platforms and AI-assisted applications using React, Spring Boot, Django, FastAPI, and modern AI tooling including Faster-Whisper, MediaPipe, and Ollama. Internship experience spans fintech, accessibility, healthcare, and educational technologies. Seeking a Summer 2026 internship in Software Engineering or Applied AI.

PROFESSIONAL EXPERIENCE

Web Developer Intern, *Esprit Tech*

Jul 2025 – Aug 2025
Tunis, Tunisia

Built HIKMALEARN: an inclusive educational assistance platform for students with disabilities, as the sole developer on both frontend and backend within a 2-month internship.

- Designed and built a full-stack accessibility platform (ReactJS frontend + Django REST backend) meeting WCAG 2.1 compliance
- Integrated cloud AI services, including automated subtitle generation for videos, screen reader support, keyboard navigation, and alt-text and image description
- Delivered all sprint milestones on schedule using Agile/Scrum methodology

Stack: ReactJS · Django · Python · REST API · Cloud AI · WCAG 2.1 · Git

Web Developer Intern, *Central Bank of Tunisia (BCT)*

Feb 2024 – May 2024
Tunis, Tunisia

Brought in to replace a manual, Excel-based event coordination process with a secure internal web platform, serving the bank's operations and logistics teams across multiple departments.

- Built internal event management web application for bank staff covering scheduling, logistics, and records management
- Implemented Spring Boot backend and Thymeleaf UI, replacing a manual coordination process with a structured digital platform
- Integrated relational SQL database ensuring secure data handling for all event records.

Stack: Spring Boot · Java · Thymeleaf · SQL · Figma · Agile

Reception Agent, *Vivre Mieux*

May 2023 – Jul 2023
Tunis, Tunisia

- Managed client communications, appointments, and social media content creation (Canva, Photoshop) in a customer-facing environment

Odoo Developer Intern, *I-Way*

Jul 2022 – Aug 2022
Tunis, Tunisia

Tasked with building a custom Odoo module to automate patient appointment scheduling, eliminating a manual booking workflow for the client's healthcare operations.

- Developed custom Odoo module for patient appointment management in Python and JavaScript, automating a previously manual booking workflow
- Analysed Odoo's technical architecture and resolved integration issues while maintaining application responsiveness.

Stack: Python · JavaScript · Odoo · PostgreSQL

Banking Intern, *BH Bank*

Jul 2021 – Aug 2021
Tunis, Tunisia

- Supported banking operations by processing client transactions and addressing client needs
- Gained hands-on exposure to banking software, financial systems, and regulated operational procedures

EDUCATION

Software Engineering — TWIN (Web & Internet Technologies), *ESPRIT School of Engineering*

2024 – Present
Tunis, Tunisia

- 2nd year Software Engineering Student specializing TWIN (Web & Internet Technologies)
- BuildDream team member: delivered 3 cross-disciplinary large-scale academic projects (AI, microservices, accessibility)
- Hextech team member: delivered 3 synchronised platforms simultaneously: Web, Mobile, and Desktop

- Foundation in CS fundamentals, multimedia, and software design
- worked on multiple ux design, 2d, and programming projects

 KEY PROJECTS

Oralis - Oral Performance Assessment Platform, ESPRIT · BuildDream • Architected and developed a full-stack AI-assisted oral assessment platform integrating React, NestJS, MongoDB, and Python-based speech analysis services. • Implemented AI pipelines for speech-to-text transcription, filler-word detection, pause analysis, pronunciation scoring, and confidence evaluation on audio/video submissions. • Designed secure and scalable backend services with JWT authentication, RBAC authorization, file upload validation, and REST APIs for students and instructors. • Built Dockerized CI/CD workflows with automated testing and Kubernetes-ready deployment pipelines. Stack: Full-Stack Web · AI/Speech Analysis · NestJs · React · FastAPI· Python	2025 – 2026
TalentLens — HR Intelligence Platform, ESPRIT · BuildDream · Machine Learning Module • Led Career Optimisation module within a 5-person academic team: built a multi-output ML pipeline using XGBoost and LightGBM to predict salary ranges and job demand from structured HR datasets • Achieved R2=0.88 on salary regression and AUC=0.75 on career classification (evaluated on held-out test set using CRISP-DM methodology) • Added SHAP-based explainability (TreeExplainer) to surface feature importance and what-if scenarios for end users • Contributed to 4 other modules: anomaly detection (Isolation Forest), market segmentation (K-Means), job recommendation (KNN) • Exposed models via FastAPI REST API; integrated with React frontend Stack: Python · XGBoost · LightGBM · scikit-learn · SHAP · FastAPI · React · Pandas · NumPy	2025 – 2026
Event Management Platform — Microservices, ESPRIT · BuildDream (6 microservices) • Owned Feedback & Reclamation microservice: full CRUD for event ratings (1-5 stars), complaints, and responses with real-time average score computation • Built temporal trend analytics and rating distribution dashboards; integrated JWT-based Auth Service (NestJS/MongoDB) for secure access control Stack: Spring Boot · Java 17 · H2 Database · REST API · JWT · Docker · Git	2025 – 2026
Pregnancy & Maternity Tracking — Multi-platform, ESPRIT • Delivered 3 synchronised platforms simultaneously: Web (Symfony/PHP), Mobile (FlutterFlow + Firebase), Desktop (JavaFX) Stack: Symfony · PHP · FlutterFlow · Firebase · JavaFX · SQL	2024 – 2025
Virtual Museum of Climate & Biodiversity, Interactive 3D Educational Experience • Built an immersive 3D virtual museum using Unreal Engine focused on climate change awareness Stack: Unreal Engine · 3D Design · Game Development	2024 – 2025

 TECHNICAL SKILLS

Languages • Python, Java, JavaScript, TypeScript, PHP, C, C++	Frameworks • React, Angular, Nestjs, Node.js, Spring Boot, Django, FastAPI, Symfony	AI / ML • scikit-learn, XGBoost, LightGBM, SHAP, TF-IDF, Pandas, NumPy, Isolation Forest, KNN, K-Means, NLP, Ollama, Whisper	DevOps • Git, GitHub, Jenkins, Docker, Kubernetes, Agile, Scrum, Postman
Databases • MySQL, PostgreSQL, MongoDB, Firebase, H2	Design / UX • Figma, Adobe XD, Photoshop, Illustrator, Canva	Mobile • FlutterFlow, JavaFX	

 LANGUAGES

Arabic	French	English Amideast B2 Certification
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